Young Innovators · Class 17

Processing: Generative Art Poster

Write the rules. Let the computer paint.

Coordinates | Loops + random | Palettes



Did you know?
Generative artworks have sold for millions, and the 'artist' wrote code, not brush strokes.

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Let's think...

Can you draw 1000 circles in 10 seconds?

Too slow by hand

A designer wants a poster full of 1000 coloured circles, all different. By hand: hours. With a **loop** and **random()**: one click, and a brand new poster every time you press Run.



Copy and paste?

Hire 10 artists?

Write a rule!

Today: you write a Processing sketch that **generates** art from rules and export it as a poster.

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Learning objectives

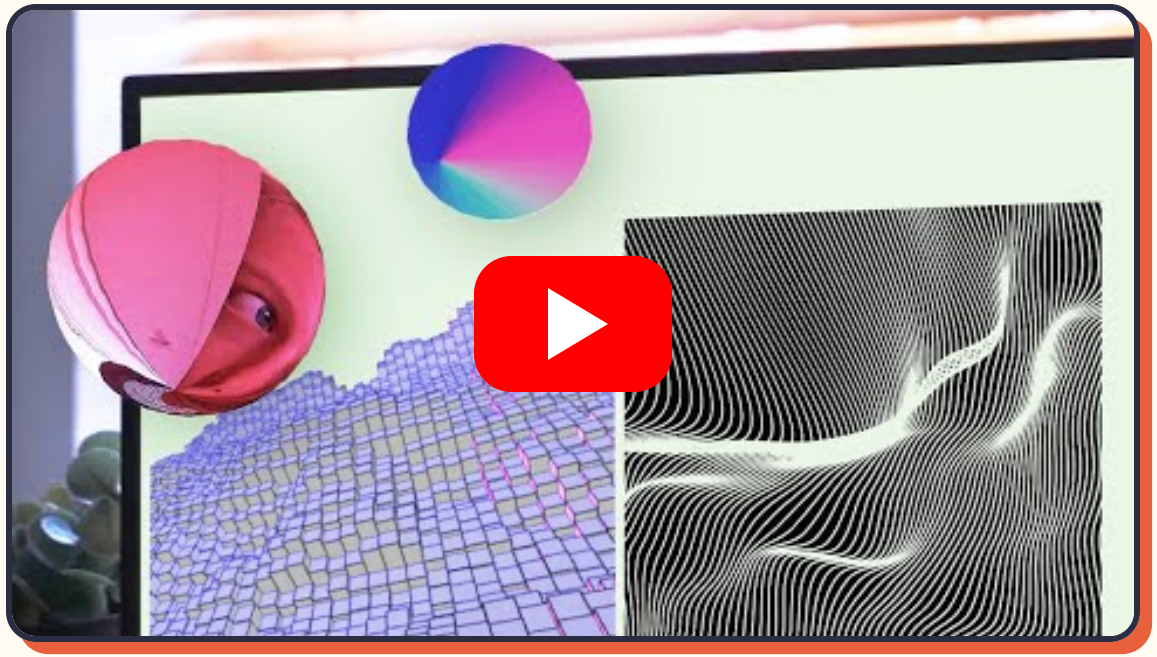
By the end of this class I can...

1	**Draw with code:** place shapes with **x, y coordinates** and colour them with **fill()**	**Proof:** I drew a planned shape exactly where I wanted it
2	**Rules + randomness:** use **loops** and **random()** to make hundreds of shapes from a few rules	**Proof:** my sketch makes a new picture every run
3	**Design a poster:** choose a **colour palette** and export a high-resolution **poster**	**Proof:** I saved a PNG poster with my name on it

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Objective 2 · Rules + randomness · video

Video time!



I Tried Making Generative Art (p5js)

phase space · about 5 minutes · youtu.be/UsIF5r8rAvk

Watch for...

1. Which **rules** did the creator write?
2. Where does **randomness** come in?
3. Which result looked best, and **why**?

Scan to watch

After the video: Name one rule you could write to make your own poster.

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Objective 1 · Draw with code

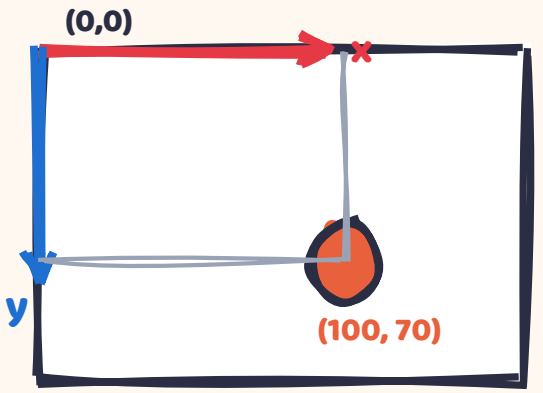
The screen is a grid of pixels

The idea

Processing draws on a grid. **(0, 0)** is the **top-left** corner. **x** grows to the right, **y** grows **downwards** (not up, like in maths!).

size(600, 800); makes a 600 × 800 canvas.
circle(100, 70, 48); draws a circle centred at (100, 70), 48 px wide.
fill(255, 120, 0); sets the colour (red, green, blue: 0 to 255).

Example: seats in a cinema



Your ticket says row 7, seat 12. Two numbers find one exact seat. Pixels work the same way, except Processing counts rows from the top.

Objective 2 · Rules + randomness

Rules + randomness = generative art

The idea

A **for loop** repeats a rule many times. **random(a, b)** picks a different number each time.

```
for (int i = 0; i < 500; i++) {
  circle(random(width), random(height), random(5, 60));
}
```

500 circles, all in different places and sizes, from 3 lines!

Example: a snowfall



Every snowflake follows the same rules (fall, drift, melt), but each one lands in a different place. Rules + a little chance make a scene that never repeats.

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Objective 3 · Design a poster

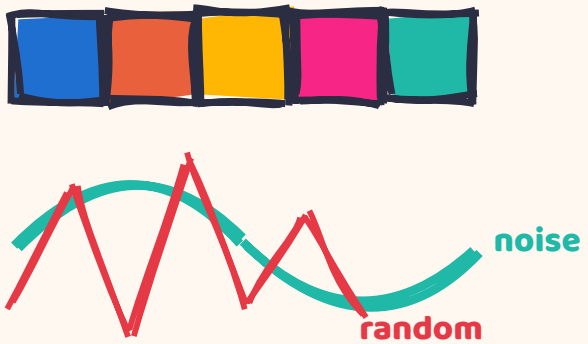
Palettes and noise make it beautiful

The idea

Pure random colours look messy. Designers pick a **palette** of 4 or 5 colours and choose from it: **color[] pal = {...};**
fill(pal[int(random(5))]);

noise() is a **smooth** random: neighbours get similar values, like hills, clouds and flowing hair. Random jumps; noise flows.

Example: truck art



Pakistani truck art uses bold repeated patterns and a strong palette. Same rules, many trucks, each one unique. That's generative design by hand!

Real world

Code that designs



Album and book covers
Many are made from generative sketches



Game worlds
Minecraft builds its landscapes with noise



Logos that change
Brands with a logo generated fresh for each use



Film effects
Smoke, fire and crowds are made by rules



Where else could you use a design that is different every time?

Same seed, same art
randomSeed(42) makes 'random' repeat exactly. Artists use it to re-create a favourite result.

Quick quiz

Read the code

Question	Answer
1. Where does circle(0, 0, 50) appear?	
2. size(400, 300): how wide is the canvas?	
3. A loop runs i = 0 while i < 200. How many shapes?	
4. fill(0, 0, 255) is which colour?	
5. Smooth waves: random() or noise()?	

Answers

Cover! 1: centred on the top-left corner · 2: 400 px · 3: 200 · 4: blue · 5: noise()

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Human art machine
2. 600 circles from one loop
3. Make it yours
4. Flowing lines with noise()
5. Export your poster
6. Debug and gallery
7. Artist challenges

MAKE

Class game

Human art machine

How to play

1. Each pair gets a blank A4 sheet and two dice.
2. **Rule:** roll for x (1 to 6 → column), roll for y (1 to 6 → row).
3. Draw a circle there. Size = (die 1 + die 2) cm. Colour = from YOUR palette of 3.
4. Repeat 15 times. Compare: same rules, different art!

Then discuss

Which part of the picture came from the rules, and which from chance?



Big idea

This is exactly what our loop does, just 1000× faster.

MAKE

Objective 2 · Rules + randomness · Build step 1 of 3

600 circles from one loop

```
void setup() {
  size(600, 800);
  background(20, 33, 61);
  noStroke();
  color[] pal = {#1F6FD1, #E8603C, #FFB703,
                 #F72585, #20B8A6};
  for (int i = 0; i < 600; i++) {
    float x = random(width);
    float y = random(height - 120);
    float d = random(6, 70);
    fill(pal[int(random(pal.length))], 200);
    circle(x, y, d);
  }
  fill(255);
  textSize(48);
  textAlign(CENTER);
  text("YOUR NAME", width/2, height - 50);
}
```

Run it

Press ▶. Press it again: a **new** poster every time!

Alpha

The 200 in fill(colour, 200) makes shapes see-through, so overlaps mix.

Try

Change 600 to 3000, or 70 to 200. What happens to the mood?

MAKE

Objective 3 · Build step 2 of 3

Make it yours

a Choose **your own palette**: 4 to 6 colours (try coolors.co for ideas).

b Swap circles for **rect()**, **triangle()** or **line()**, or mix them.

c Add a rule: shapes get **smaller** near the bottom: d = random(5, 80) * (1 - y/height).

d Add **randomSeed(**your birthday**)** so you can recreate a favourite.

e Put your name and a title in the text.



Check: Your poster has its own palette and at least one extra rule.

MAKE

Objective 2 · Rules + randomness · Level up: noise lines

Flowing lines with noise()

```
float t = 0;
void setup() {
  size(600, 800);
  background(250, 245, 235);
  stroke(232, 96, 60, 40);
  noFill();
}

void draw() {
  beginShape();
  for (int x = 0; x <= width; x += 10) {
    float y = noise(x * 0.004, t) * height;
    vertex(x, y);
  }
  endShape();
  t += 0.004;
  if (frameCount > 400) noLoop();
}

void keyPressed() {
  if (key == 's') save("poster.png");
}
```

draw()

draw() runs 60 times a second. Each frame adds one wavy line; 400 lines build silk-like layers.

noise

Small numbers like 0.004 = smooth, gentle waves. Try 0.02 for wild ones.

Save

Press **s** to save poster.png into the sketch folder (Sketch → Show Sketch Folder).

MAKE

Objective 3 · Build step 3 of 3

Export your poster

a Pick your best sketch (circles, lines, or both combined).

b For print quality, add **pixelDensity(2);** under size(): the image is twice as sharp.

c Press **s** (or add **save("poster.png");** at the end of setup()).

d Open the PNG. Check your name is visible and nothing is cut off.

e Upload to the class folder for the gallery.



Check: A PNG poster with your name is in the class folder.

MAKE

Test like an engineer

Debug and gallery

Check	Yes / No	Notes
Sketch runs without red errors		
New picture on every run		
Uses my own palette		
Has at least one extra rule		
Poster PNG saved with name		

Debug checklist

Nothing appears? Did you call background() inside draw()? It wipes the canvas every frame.
Error 'missing semicolon'? Check the line ABOVE the red one.

Gallery walk: stand by your poster and explain one rule you wrote.

MAKE

Level up!

Artist challenges

Choose your challenge



Interactive: use mouseX and mouseY to change colours or sizes live.



Grid: use two nested loops to make a tiled pattern with random rotations.



Truck art: design a repeating border inspired by Pakistani truck art.



Animate: make the shapes slowly drift with noise and record a GIF.

Think like an engineer

Who is the artist: you, or the computer?

Boss level

Make a series of 3 posters with the same rules but different seeds, and print the best one.

Objective check

Can I really do it? Prove it.

01	Where is (0, 0) in Processing, and which way does y grow?	○ ○ ○
02	Show the loop and random() calls in your sketch. random() vs noise()?	○ ○ ○
03	Show your exported poster PNG and name your palette.	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Print the class's best posters for the school corridor and add a QR code to each sketch.

Show what you know

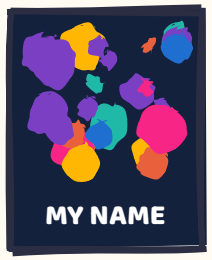
What did we learn?

- [x] (0, 0) is the top-left; y grows downwards
- [x] Loops repeat a rule many times
- [x] random() jumps; noise() flows smoothly
- [x] A small palette makes art look designed
- [x] save() exports your poster

Exit ticket: how many shapes does 'for (int i = 0; i < 50; i++)' draw?

At home

Open a book cover or poster at home. Could a rule have made it?



Next class: Processing + Arduino: Live Sensor Radar!